

Course Syllabus

Course	EMEN7029 & EMEN7104 Fall 2026
Time	19:00-22:00, Fridays
Location	Room E12-G004
Description	This course introduces the theory of deep neural networks and its application in robotics
Instructor	Qingbiao Li <qingbiaoli at um.edu.mo>
Office Hours	10:00-12:00 Friday in N21-2012g or E11-4065
Grading	• Assignments: 50% • Final Project: 40% • Participation: 10%
Cheating Policy	Any students caught cheating will fail the entire course
Late Work Policy	• -25% 0-1 days late • -50% 1-2 days late • -75% 2-3 days late • -100% 3+ days late
GenAI Policy	• The use of GenAI is permitted within the course policy. • Where GenAI use is permitted, students are required to acknowledge the AI tools used and explain how they were used in their assignment submissions.
Prerequisites	• Linear Algebra • Multivariable Calculus • Programming in Python
Preliminary Lecture Schedule	• Week 1 (08.21): Course Introduction • Week 2 (08.28): Linear Regression (D2L 3.1, 3.6) • Week 3 (09.04): Invited talk and Case Study for Robot Learning • Week 4 (09.11): Neural Networks (D2L 5.1, 5.2, 6.1) • Week 5 (09.18): Backpropagation and Optimization (D2L 5.3, 12.1, 12.3-12.5) • Week 6 (09.25): Classification (D2L 4.1, 4.2, 4.4) • Week 7 (10.02): Public Holiday for The Day following National Day of the People's Republic of China • Week 8 (10.9): Training Tricks (D2L 5.1-5.5, 6.1-6.3, 12.1-12.10) • Week 9 (10.16): Convolutional Neural Networks (D2L 7) • Week 10 (10.23): Recurrent Neural Networks (D2L 9-10) • Week 11 (10.30): Autoencoders and Generative Models • Week 12 (11.06): Diffusion Models for Robot Learning • Week 13 (11.13): Attention and Transformers (D2L 11.1-11.7) • Week 14 (11.20): Imitation Learning for Robotics • Week 15 (11.27): Foundation Models for Robotics (D2L 11.8-11.9)

Preliminary Assignment Schedule	• Week 1-2 (08.21 - 08.28): (Optional) Programming Practice • Week 2-4 (08.28 - 09.11): Linear Regression • Week 4-6 (09.11 - 09.25): Neural Networks and Backpropagation • Week 6-8 (09.25 - 10.9): MLP Regression • Week 8-10 (10.10 - 10.23): Convolutional NN for Object Classification based on Robot Vision • Week 10-12 (10.23 - 11.06): RNN Robot Trajectory Prediction • Week 11-12 (10.30 - 11.06): Final Project Plan • Week 12-16 (11.06 - 12.05): Final Project
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